

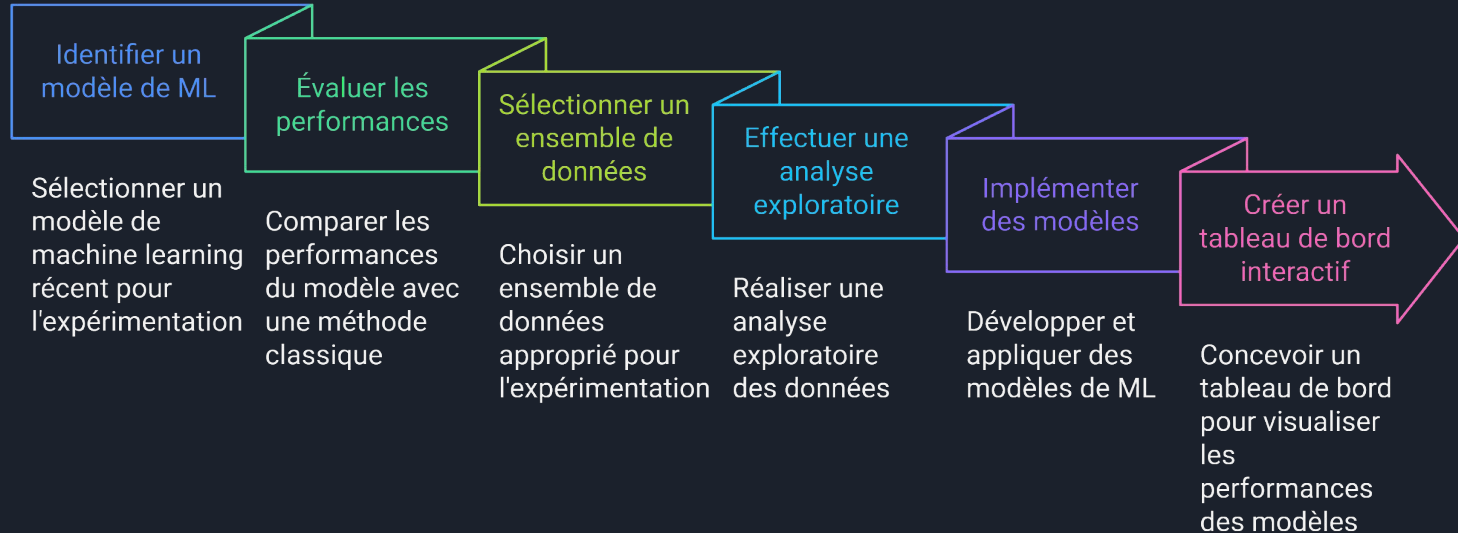
A blue parallelogram and a light green parallelogram are positioned on the left side of the slide, overlapping each other and the dark blue background.

# Développez une preuve de concept

Projet 9 - Julien Agneray

# Contexte et objectifs

## Veille technologique et approche innovante





# Sommaire

1. Présentation du Dataset
2. Choix des modèles : Chronos, ARIMA et AutoGluon
3. Méthodologie et références utilisées
4. Comprendre les modèles et leur fonctionnement
5. Évaluation et comparaison des modèles
6. Démonstration du dashboard
7. Conclusion

# Présentation du Dataset

Nom : Truck Sales for Time Series

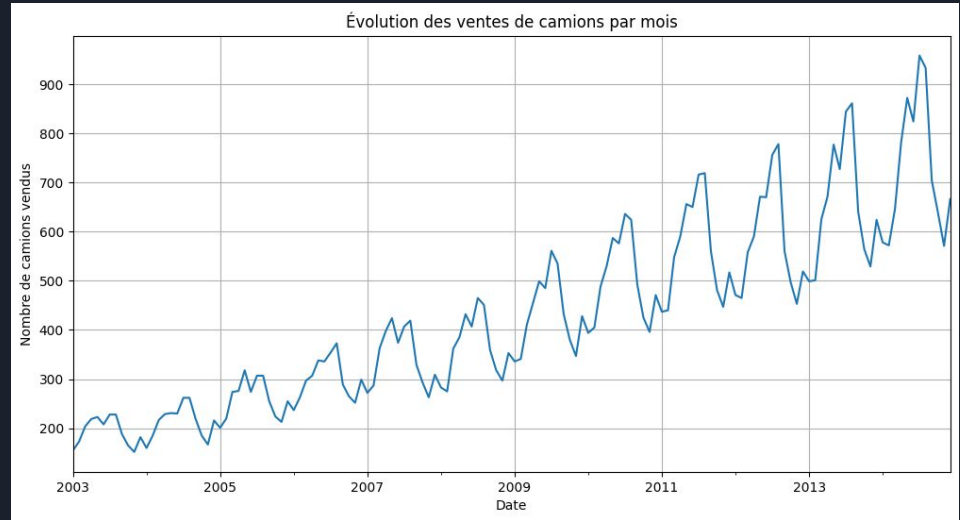
Période couverte : Janvier 2003 -  
Décembre 2014

Variables :

1. Date (mois-année)
2. Nombre de camions vendus

Avantages :

1. Données propres, sans valeurs manquantes
2. Période étendue pour analyser les tendances
3. Complexité adaptée pour la comparaison de modèles



# Pourquoi Chronos ?

# CHRONOS

Cornell University

We gratefully acknowledge support from the Simons Foundation, member institutions, and all contributors. [Donate](#)

arXiv > cs > arXiv:2403.07815

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## Computer Science > Machine Learning

[Submitted on 12 Mar 2024 (v1); last revised 4 Nov 2024 (this version, v3)]

### Chronos: Learning the Language of Time Series

Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Aleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, Jasper Zschiesner, Danielle C. Maddix, Hao Wang, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, Yuyang Wang

We introduce Chronos, a simple yet effective framework for pretrained probabilistic time series models. Chronos tokenizes time series values using scaling and quantization into a fixed vocabulary and trains existing transformer-based language model architectures on these tokenized time series via the cross-entropy loss. We pretrained Chronos models based on the TS family (ranging from 20M to 710M parameters) on a large collection of publicly available datasets, complemented by a synthetic dataset that we generated via Gaussian processes to improve generalization. In a comprehensive benchmark consisting of 42 datasets, and comprising both classical local models and deep learning methods, we show that Chronos models: (a) significantly outperform other methods on datasets that were part of the training corpus; and (b) have comparable and occasionally superior zero-shot performance on new datasets, relative to methods that were trained specifically on them. Our results demonstrate that Chronos models can leverage time series data from diverse domains to improve zero-shot accuracy on unseen forecasting tasks, positioning pretrained models as a viable tool to greatly simplify forecasting pipelines.

Comments: Code and model checkpoints available at this <https://url>

Subjects: Machine Learning (cs.LG); Artificial Intelligence (cs.AI)

Cite as: arXiv:2403.07815 [cs.LG]

(or arXiv:2403.07815v3 [cs.LG] for this version)

<https://doi.org/10.48550/arXiv.2403.07815>

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cs.AI

### References & Citations

- NASA ADS
- Google Scholar
- Semantic Scholar

1 blog link (what is this?)

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## amazon's Collections

Chronos-Bolt [Models](#) Chronos Models & Datasets

### Chronos Models & Datasets

updated Nov 26, 2024

Collection of artifacts related to Chronos pretrained models for time series forecasting.

[▲ Upvote 39](#) 

amazon/chronos-bolt-tiny

Time Series Forecasting • Updated Nov 28, 2024 •  $\pm 171k$  •  $\heartsuit 13$

amazon/chronos-bolt-mini

Time Series Forecasting • Updated Nov 28, 2024 •  $\pm 11.5k$  •  $\heartsuit 5$

amazon/chronos-bolt-small

Time Series Forecasting • Updated Nov 28, 2024 •  $\pm 2.8M$  •  $\heartsuit 7$

amazon/chronos-bolt-base

Time Series Forecasting • Updated Nov 28, 2024 •  $\pm 573k$  •  $\heartsuit 26$

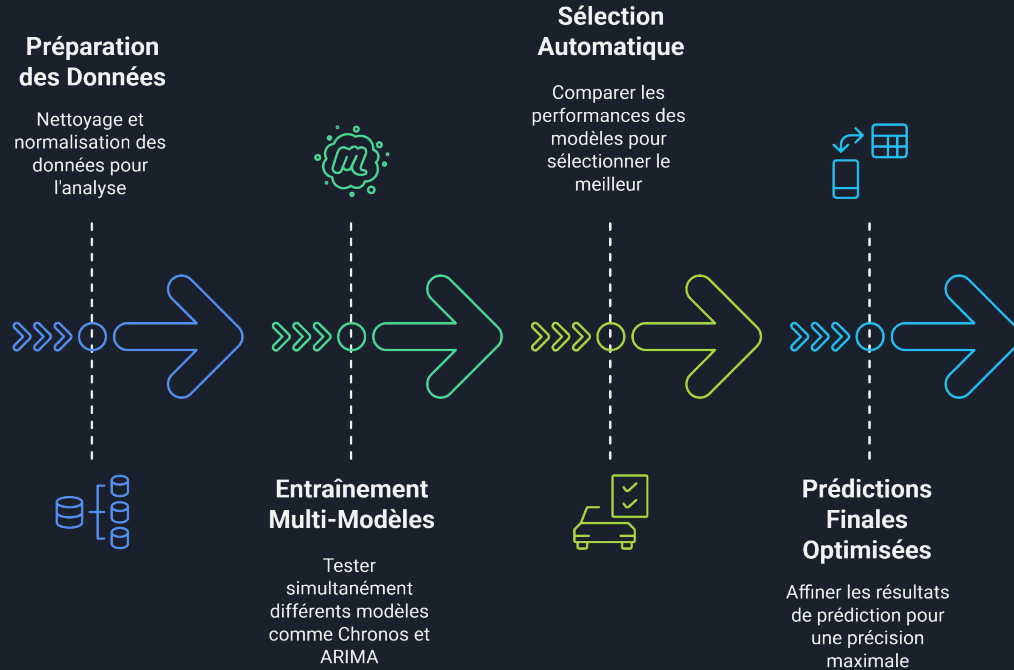
amazon/chronos-t5-tiny

Time Series Forecasting • Updated Nov 27, 2024 •  $\pm 448k$  •  $\heartsuit 104$

# ARIMA - Le modèle de référence



# AutoGluon - Automatiser la sélection du meilleur modèle





# Sources et références utilisées

## Publications scientifiques

- ♦ Chronos: Learning the Language of Time Series (Arxiv, 2024)  
<https://arxiv.org/abs/2403.07815>

## Frameworks et documentation

- ♦ Chronos Forecasting - GitHub (Amazon Science)  
<https://github.com/amazon-science/chronos-forecasting>

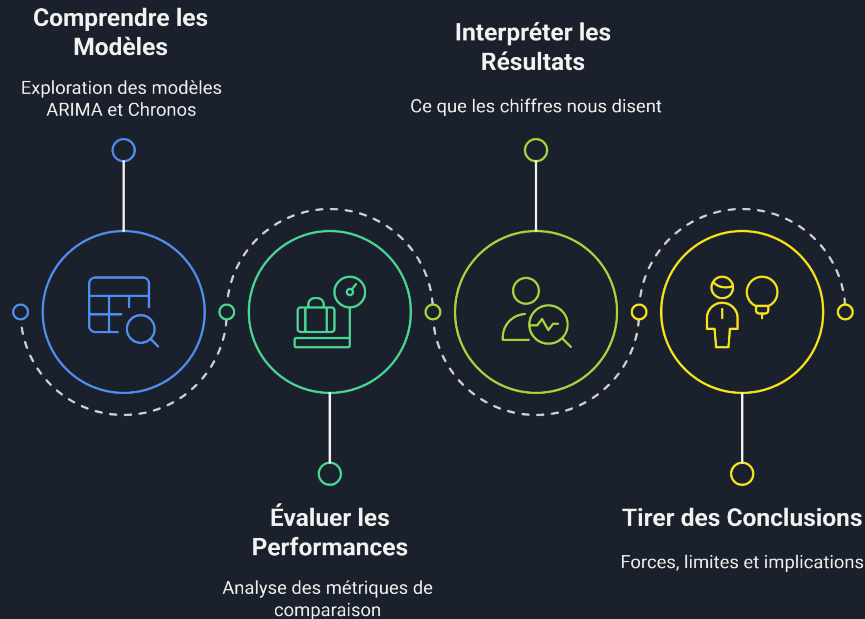
## Analyses et veille technologique

- ♦ Paper Review: Chronos - Lukyanenko (2024)  
<https://andlukyane.com/blog/paper-review-chronos>



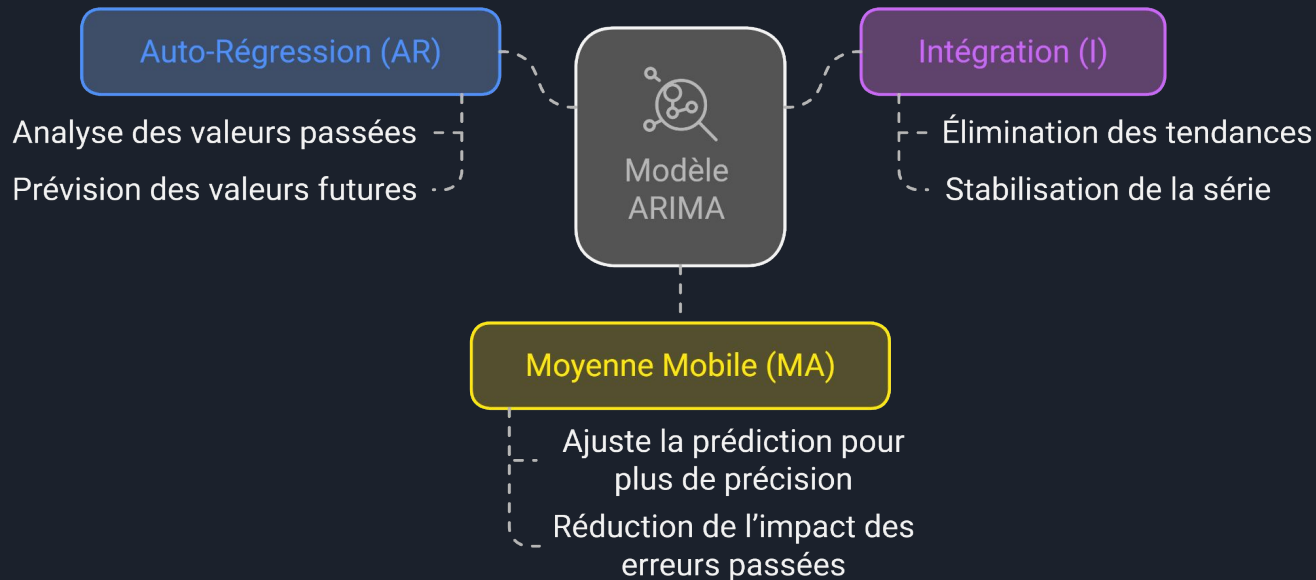
# De la théorie aux résultats – Une approche rigoureuse

Analyse et Interprétation des Modèles ARIMA et Chronos



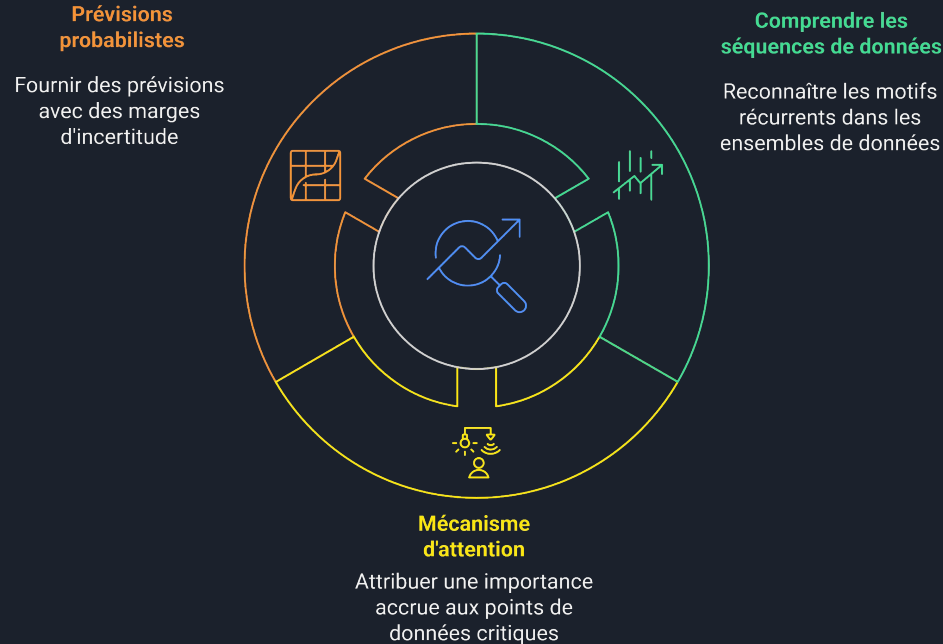
# Comment fonctionne ARIMA ?

## Modèle ARIMA: Composantes et Fonctions

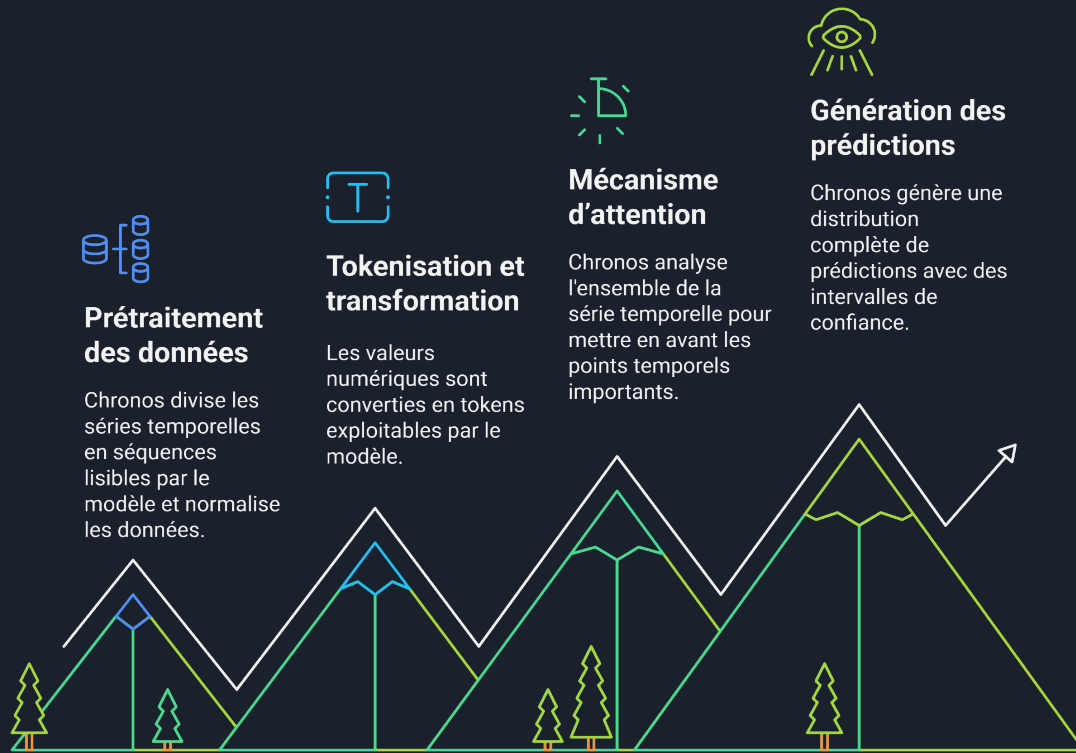


# Comment Chronos aborde les séries temporelles ?

Chronos : Comprendre et prédire les séries temporelles

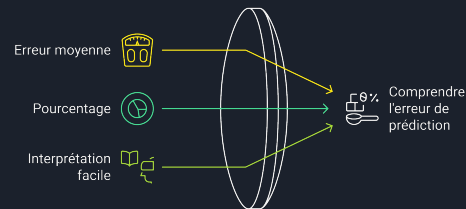


# Étapes clés du fonctionnement de Chronos

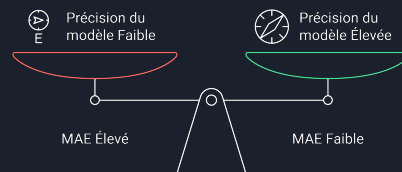


# Évaluation des modèles

MAPE => Erreur moyenne en pourcentage



MAE => Erreur absolue moyenne



RMSE => Erreur quadratique moyenne



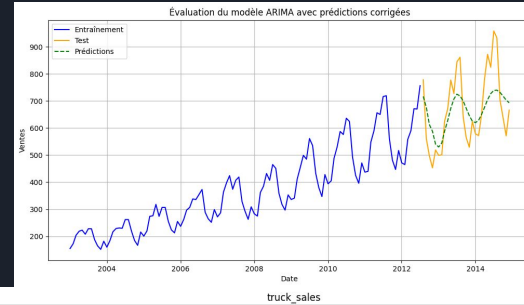


## Résultats des tests – Différences entre manuel et AutoGluon

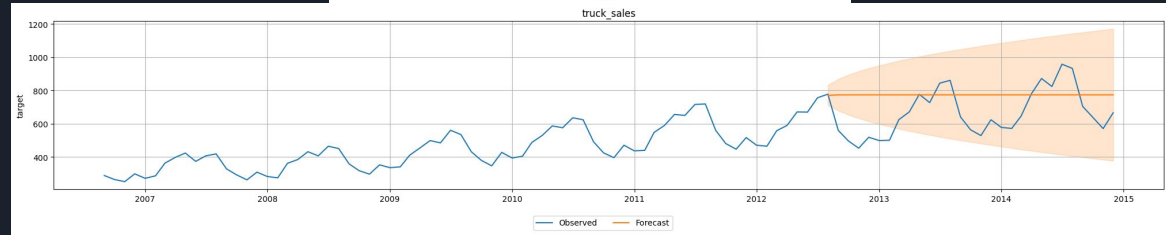
| Modèle  | Test            | MAPE (%) | MAE    | RMSE   |
|---------|-----------------|----------|--------|--------|
| ARIMA   | Test individuel | 12.39    | 83.56  | 100.67 |
| ARIMA   | AutoGluon       | 25.63    | 150.26 | 173.68 |
| Chronos | Test individuel | 6.11     | 40.63  | 48.22  |
| Chronos | AutoGluon       | 6.11     | 40.63  | 48.22  |

# Analyse des performances et conclusions

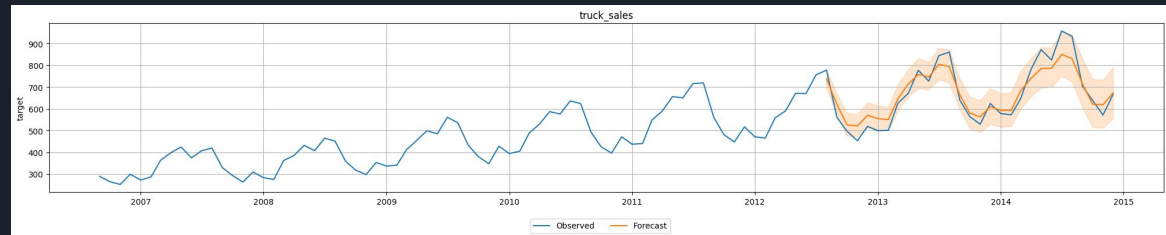
ARIMA - Paramétrage manuel



ARIMA - AutoGluon



Chronos - AutoGluon



# L'importance des variables dans la prédiction

## Impact des Variables sur la Prédiction

### Modèle à Variable Unique



Se concentre sur une seule variable sans relations complexes



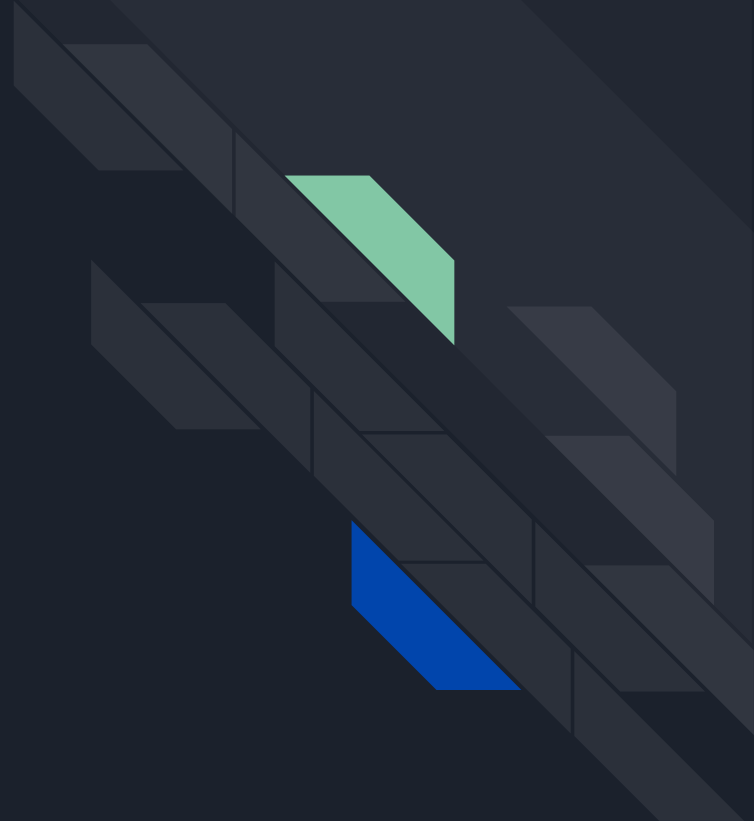
### Modèle à Variables Multiples

Exploite des relations complexes entre les variables (Météo, Économie, Promotions)



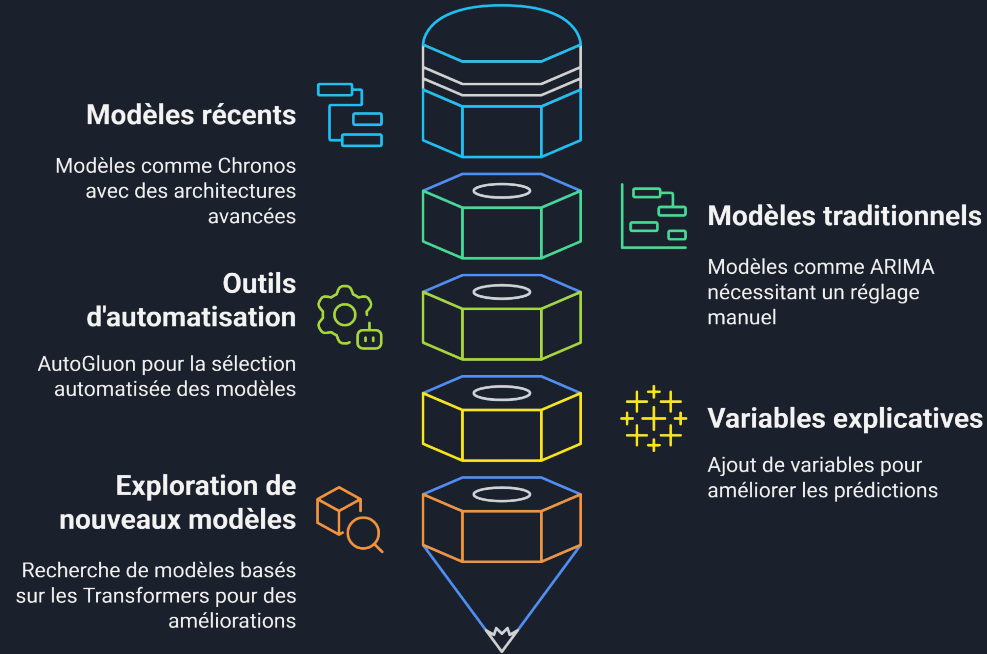


# Démonstration du dashboard



# Conclusion

## Amélioration des prévisions de séries temporelles



Avez-vous des questions?

